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ABSTRACT

IR for AAC Users: A Hyperdimensional Computing (Vector Symbolic Architectures) Approach

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Open Access Support provided by:

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Published: 13 July 2025

[Citation in BibTeX format](#)

SIGIR '25: The 48th International ACM
SIGIR Conference on Research and
Development in Information Retrieval
July 13 - 18, 2025
Padua, Italy

Conference Sponsors:
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IR for AAC Users: A Hyperdimensional Computing (Vector Symbolic Architectures) Approach

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Abstract

This work proposes Hyperdimensional Computing (HDC) as a design paradigm [13] to facilitate search and recommendation activities for disabled users employing symbolic augmentative and alternative communication (AAC) systems. Such a context necessitates flexibility and composability in item and query representations as a consequence of vocabularies being tailored to an individual user. HDC is suggested to meet these needs in an efficient manner. However, construction and empirical evaluation are left to additional research.

Project development for symbolic AAC supports a small but highly diverse user base. This creates a “low resource environment” from both an economic and technical perspective. The heterogeneity of user communication abilities, preferences, and symbolic interpretation limits the generalisability of datasets and presents a cold start problem. Furthermore, countries with underdeveloped social welfare and health infrastructure, the Philippines serving as an illustrative example, have few practitioners able to fine-tune AAC devices or support inclusive decision making. Consequently, this exacerbates resource constraints, which motivates systems to offer consumer-level ease of use. Rural areas, in any socioeconomic setting, lack reliable network connectivity: designers should anticipate regular on-device computation, leaving networked tasks to limited occasions. Technologies for disabilities, broadly speaking, are related to all 17 UN Sustainable Development Goals [21]. Additionally, AAC is a focus area of UNICEF’s work on disabilities and inclusion for marginalised children [2].

By proposing HDC as a way to mediate diverse feature spaces, search and recommendation applications are able to reuse semantic components and process structures — reducing costs. This flexibility enables adaptation to underserved cultures, where localised symbol interpretations are particularly needed [3]. A similar approach has been taken to circulate open access symbols [2]. HDC’s development comes from an interdisciplinary perspective, implementing cognitive and linguistic science precepts computationally [8]. Building on this idea, AAC for search requires wider collaboration with entities outside the traditional information retrieval (IR) community, such as speech pathologists, disability support workers, and end users with disabilities.

ACM Reference Format:

Hunter Briegel, Maya Pagal, and J. Shane Culpepper. 2025. IR for AAC Users: A Hyperdimensional Computing (Vector Symbolic Architectures) Approach. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '25)*, July 13–18, 2025, Padua, Italy. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3726302.3730273>

1 Context

Certain groups of disabled people rely on AAC to express themselves. Although this can encompass a wide range of techniques, a pictorial or symbolic vocabulary is frequently deployed for users with linguistic and literacy deficits [23]. These representations relate either to words or higher-level concepts. When considering the importance of personal autonomy in decision making, IR is an activity that directly enables choice and control for disabled people in their daily lives. The UN Convention on the Rights of Persons with Disabilities [24] promotes personal decision making, alongside universal design and accessibility [18].

Previous work that combines IR with AAC has primarily focused on approaches that retrieve longer utterances to improve a user’s speech efficiency [15, 17]. IR can also be important when symbolic AAC is used directly as a query vocabulary. That is, using AAC input sources for search and recommendation over diverse data [20]. In particular, users with cognitive impairment, developmental delays, or brain damage [4] might benefit from this approach as text completion is not sufficient, impeding access to mainstream information services.

Given recent reforms to state support systems, beneficiary participants are empowered with greater choice and control over personal disability budgets [19]. For instance, they can self-select assistive technology (AT) [6], support workers, and therapeutic activities. In order to fully realise the goals of these reforms, the beneficiary must have as much autonomy as reasonably possible in determining their own care. Being able to independently query an IR system is one technical means of enabling this objective.

In daily settings, recommendation can also be important when exercising autonomy. For example, an IR system can be used to recommend activities or food, among other things. It is envisaged that by linking a familiar symbolic vocabulary to a data repository, a user with limited communication abilities could better express their information needs over real-world alternatives. A system in this context would enable best practices in supported decision making, a concept where disabled persons are assisted to make choices with trusted associates [5].

Input may be sourced from symbols presented on a dedicated device, personal computer, phone, or tablet possessing varying system resources. In the Philippines, several groups have developed

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SIGIR '25, Padua, Italy

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ACM ISBN 979-8-4007-1592-1/2025/07

<https://doi.org/10.1145/3726302.3730273>

localised applications for AAC communication [1]. Notably, these leverage consumer hardware to reduce access barriers.

1.1 Personalisation

In order to connect with a user’s personal experience and real-world contexts, AAC interfaces are often highly personalised. In modern applications, photographs can be uploaded to represent familiar scenes [23]. These trigger a connection for people with memory loss or cognitive impairments. As the meaning of these objects (photos or symbols) can be ambiguous, a family member or health care provider is needed to help define key features related to the experience. These may be manually composed of discrete basis features or generated using sentence embeddings of natural language descriptions [7]. A particular photograph can have multiple facets or dimensions related to a search task. For example, a restaurant photograph could remind the system user of the physical environment, others present, or the type of food served. When mapped to an information need, past photographic reminders can be used to search for similar items. By binding an image with a key representing a facet, also expressed through symbolic means, the desired meaning can be translated to the search engine.

Furthermore, images can be combined with other symbols to search by analogy. For instance, one could seek foods similar to a reference image of enjoying pasta in other cuisines. Here, the image is bound to a role representing the cuisine as opposed to a specific restaurant environment or social milieu.

2 Suggested Methodology

Hyperdimensional Computing, also known as Vector Symbolic Architectures [8, 12], refers to the use of high-dimensional vectors to represent and process objects using simple mathematical operators [12, 22]. The methodology meets the requirements of the AAC domain: in particular, resource-constrained devices and environments, alongside personalisation.

As a basis representation, HDC leverages the high dimensionality of randomly initialised vectors to create approximately orthogonal relations in a geometric space [12]. However, when these vectors are used as building blocks, they can be aligned with related concepts. This capability allows a mapping of personalised AAC symbolic vocabularies to a common semantic featurisation of alternatives. In the search space, items can be assigned descriptive features using an LLM, a classifier model, or with human annotation. Additionally, pretrained features from other models can be projected into the hyperdimensional space [7, 16]. For example, an activity can be classified by location, therapeutic contribution, or social elements. These feature vectors, and their temporal or causal sequences, can then be used directly or associated with a familiar representation. AAC users then complete queries using such representations without needing to understand the vectorised subcomponents.

Linking unique vectors is achieved through binding, bundling, and permutation. The first operation, binding, creates a key-value relationship using element-wise multiplication. For instance, “outdoors” could be a location type. A bound vector is therefore:

$$\text{relationship} = \text{location} \odot \text{outdoors} \quad (1)$$

Bundling combines vectors together element-wise to represent a set of query attributes or an item/document to retrieve. An example bundled hypervector representation of an activity could be:

$$\text{item} = \text{location} \oplus \text{action} \oplus \text{equipment} \oplus \text{skills} \quad (2)$$

A hash-table encoding combines the two above operations [25]:

$$\bigoplus_{i=1}^n K_i \odot V_i \quad (\text{See Vergés et al.}) \quad (3)$$

Permutation enables sequence ordering of component hypervectors to be created. Other data structures and embedding types are possible, and are supported by open-source libraries [10]. These extensions enable more complex types, such as levels or feature transformation [14].

A supporter who is familiar with the user can determine the meaning of the available symbols and create semantic vectors without requiring any retraining. These linkage actions enable complex arrangements of available taxonomy features to be created for a set of alternatives. This is a key motivator when adopting HDC, as an open or novel vocabulary is rarely suitable for AAC users with cognitive or memory impairment. The following exemplar query instantiates “cardio” as a familiar conceptual element:

$$\text{cardio} = \text{act} \odot (\text{running} \oplus \text{cycling}) \oplus \text{loc} \odot (\text{indoors} \oplus \text{outdoors}) \quad (4)$$

$$\text{query} = \text{cardio} \oplus (\text{skills} \odot \text{mobility}) \quad (5)$$

During retrieval, a distance similarity measure, such as cosine similarity, can be used to locate items matching the query. HDC also enables efficient on-device training to refine recommendations or to predict intent [17]. Known techniques in the literature require a single pass over the data and do not use gradient descent or neural networks to build the final model [9]. Additional research has improved model accuracy using more complex learning approaches [11]. One can imagine a supporter or user providing ratings for known items or previously undertaken actions. The attributes of these elements can be used to train a re-ranker to prioritise the top- k results returned based on past user preferences through a secondary similarity comparison. Contextual information could further improve recommendations through the same process.

3 Future directions

Our contribution suggests HDC as a well-aligned computing framework for improving the accessibility of search and recommendation. To ensure a successful real-world implementation, additional research is needed. A developer needs to understand which capabilities of HDC to personalise symbolic vocabulary are preferred by users. The selection of HDC as opposed to alternative approaches should be assessed, including from an efficiency point of view. Lastly, regardless of retrieval technology, the presentation of search results in an interpretable manner or their inclusion in supported decision making is also an important open research problem.

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